Task 1: Real Router Image and Port Identification

The real router photo below isfromWikimedia Commons. It shows therear ofaTP-LinkArcher A6 router, where the WAN and LAN ports, power jack and buttons can be seen. The LED layout varies by model, so check the front panel for status lights.

Labels to identify on the photo: WAN port — connection to the modem/ONT; LAN ports — wired computers or a switch; power jack — supplies electricity; WPS/Wi-Fi and Reset buttons — wireless setup and reset functions. Status LEDs are usually on the front/top of the router.

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Task 2: Three key differences between routers and switches

Point	Router	Switch
Main job	Connects different IP networks, such as a home LAN and the Internet.	Connects devices inside the same local network (LAN).
Address used	Uses IP addresses and routing information to choose where packets go next.	A typical Layer 2 switch learns MAC addresses and forwards Ethernet frames to the right port.
Traffic handling	Chooses a path between networks; home models often provide NAT, DHCP and firewall features.	Forwards local traffic efficiently without sending every frame to every port.

Easy way to remember: the router connects networks; the switch connects devices inside a network. When I open a website, the router sends traffic from my home network toward the Internet. When my laptop sends a file to a printer on thesame LAN,aswitch can forward that traffic locally.

Task 3: Identify ports, power and LED indicators

On areal device, theexact position depends onthe brand andmodel.The power port is usually marked “Power” or uses a round DC jack. Ethernet ports are wider sockets labelled WAN, Internet or LAN. LEDs may be next to each port or on the front. A steady light often means a link is present, while blinking can indicate activity; the device manual is the final guide.

Router: Power input supplies electricity. WAN/Internet connects to the ISP modem/ONT. LAN ports connect a PC, TV or switch. Status LEDs show power, Internet/WAN, Wi-Fi and LAN link/activity. A console port is usually found only on business/managed models.

Switch: Power input supplies electricity (some switches can receive power over Ethernet). Ethernet ports connect PCs, printers, access points or a router. Port LEDs usually indicate link and activity; managed switches may also have a console port.

Task 4: Small gaming café network scenario

If I weresetting up a small gaming caféfor IPLstreamingand multiplayer matches, I would connect the ISP line to the modem/ONT first, then connect the modem/ONT to the router’s WAN port. After that, I would connect one LAN port on the router to an Ethernet switch. The eight gaming PCs would plug into eight switch ports, using Ethernet cables for a more stable connection than Wi-Fi.

I would keep the router between the café network and the Internet because it handles IP routing and usually provides DHCP and firewall protection. The switch is the practical choice for the eight PCs because it gives them wired ports and forwards local traffic using MAC addresses. For smooth IPL streaming, I would also avoid overloading the Internet connection and use good-quality cables.